

Numerical Analysis PhD Exam, January 2010

Answer any 8 questions.

1. Prove that a projector is normal if and only if it is self-adjoint.
2. Let $A \in \mathbb{R}^{N \times N}$ and $b \in \mathbb{R}^N$. Consider the following iteration for solving $Ax = b$, that computes x_{n+1} , given $x_n \in \mathbb{R}^N$, as follows (in m intermediate steps): Setting $x_{n+1}^{(0)} = x_n$, for $\ell = 1, 2, \dots, m$, compute $x_{n+1}^{(\ell)} = x_{n+1}^{(\ell-1)} + \tau_\ell(b - Ax_{n+1}^{(\ell-1)})$. Then, define $x_{n+1} = x_{n+1}^{(m)}$. There is a linear operator E such that $x_{n+1} - x = E(x_n - x)$. Give a formula for E . Suppose A is Hermitian and positive definite with spectral condition number κ . Prove that there are real values of the m parameters τ_ℓ such that

$$\rho(E) \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^m.$$

3. Suppose A and B are square matrices such that AB is normal. Prove that $\|AB\|_2 \leq \|BA\|$. (We use $\|\cdot\|_2$ to denote the spectral norm and $\|\cdot\|$ to denote any induced matrix norm.)
4. Let $A \in \mathbb{C}^{m \times m}$, and a_j be its j -th column. Prove that

$$|\det A| \leq \prod_{j=1}^m \|a_j\|_2.$$

5. Suppose A is a Hermitian positive definite matrix split into $A = C + C^* + D$ where D is also Hermitian positive definite. Show that $B = C + \omega^{-1}D$ is invertible. Consider the iteration $x_{n+1} = x_n + B^{-1}(b - Ax_n)$, with any initial iterate x_0 . Prove that x_n converges to $x = A^{-1}b$ whenever $0 < \omega < 2$.
6. Let $x_0, x_1, \dots, x_n$ be distinct points in a finite interval $[a, b]$ and $f \in C^1[a, b]$. Show that for any given $\varepsilon > 0$ there exists a polynomial p such that $\|f - p\|_\infty < \varepsilon$ and $p(x_i) = f(x_i)$ for all $i = 0, 1, \dots, n$ (where $\|\cdot\|_\infty$ denotes the $L^\infty(a, b)$ -norm).
7. Let $p > 0$ and $x = \sqrt{p + \sqrt{p + \sqrt{p + \dots}}}$, where all the square roots are positive. Design a fixed point iteration $x_{n+1} = F(x_n)$ that converges to x . You should write down a specific F . Obtain a sufficient condition on the initial iterates for (global) convergence of the iteration.
8. Let n be a positive integer, $h = 1/n$, and consider the grid of points (ih, jh) for $i, j = 0, 1, \dots, n$. Let A be the finite difference operator with the “5-point stencil” discretizing the Laplace operator $-\Delta$, with zero Dirichlet boundary conditions on this grid. Describe it. Prove that the spectrum of A consists of the numbers

$$\lambda_{lm} = 4h^{-2} \left(\sin^2(l\pi h/2) + \sin^2(m\pi h/2) \right),$$

for all $l, m = 1, \dots, n-1$.

9. Let x_m and x_{m+1} be two successive (complex) iterates when Newton's method is applied to a polynomial $p(z)$ of degree n . Prove that there is a zero of $p(z)$ in the disk $\{z \in \mathbb{C} : |z - x_m| \leq n|x_{m+1} - x_m|\}$.
10. Let $w(x)$ be a positive integrable function on $[a, b]$. Consider the quadrature

$$\int_a^b f(x)w(x) dx \approx \sum_{k=0}^n \left(A_k f(x_k) + B_k f'(x_k) + C_k f''(x_k) \right)$$

for any f with continuous first and second derivatives (f' and f'' , respectively). Give conditions on A_k, B_k, C_k and x_k so that the quadrature has precision $4n + 3$.